

Yulu Pan

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SUMMARY & RESEARCH INTEREST

Ph.D. student in Computer Science at UNC-Chapel Hill researching **computer vision**, **video understanding**, and **AI for sports**. I build large-scale video datasets and benchmarks that move models from fine-grained skill recognition toward causal, strategic, and agentic reasoning in long videos.

EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

Ph.D. in Computer Science; Advisor: [Prof. Gedas Bertasius](#)

Aug. 2025 – Present

M.S. in Computer Science

May 2025

B.S. in Computer Science and B.A. in Mathematics, GPA: 3.74 / 4.0

May 2023

PUBLICATIONS

- *Ce Zhang, Ziyang Wang, [Yulu Pan](#), Oluwatumininu Oguntola, Pranav Wagh, Qiyu Wu, Hiromi Wakaki, Mohit Bansal, and Gedas Bertasius.*
Searching Videos as Trees: Self-Correcting Agents for Grounded Long Video QA. (**arXiv 2026**) [\[link\]](#)
- *[Yulu Pan](#), Han Yi, Seongsu Ha, Md Mohaiminul Islam, Benjamin Zhang, Lorenzo Torresani, and Gedas Bertasius.*
SVI-Bench: A Dynamic Microworld for Strategic Video Intelligence. (**ECCV 2026**) [\[link\]](#)
- *Han Yi, [Yulu Pan](#), Feihong He, Xinyu Liu, Benjamin Zhang, Oluwatumininu Oguntola, and Gedas Bertasius.*
ExAct: A Video-Language Benchmark for Expert Action Analysis. (**NeurIPS 2025, Datasets and Benchmarks Track**) [\[link\]](#)
- *[Yulu Pan](#), Ce Zhang, and Gedas Bertasius.*
BASKET: A Large-Scale Video Dataset for Fine-Grained Skill Estimation. (**CVPR 2025**) [\[link\]](#)
- *Akshay Paruchuri, Xin Liu, [Yulu Pan](#), Shwetak Patel, Daniel McDuff, and Soumyadip Sengupta.*
Motion Matters: Neural Motion Transfer for Better Camera Physiological Sensing. (**WACV 2024, Oral**) [\[link\]](#)
- *Jianjian Yin, Zhichao Zheng, [Yulu Pan](#), Yanhui Gu, and Yi Chen.*
Semi-Supervised Semantic Segmentation with Multi-Reliability and Multi-Level Feature Augmentation. (**Expert Systems with Applications, Volume 233, 15 December 2023, 120973**)

EXPERIENCE

Graduate Research Assistant

Dec. 2023 – Present

UNC-Chapel Hill, Advisor: [Prof. Gedas Bertasius](#)

Chapel Hill, NC

- Developed [SVI-Bench](#), a 35,000-hour multi-sport benchmark with 15M annotated actions and nine tasks across basketball, soccer, and hockey. (**ECCV 2026**)
- Curated BASKET: 4,400+ hours and 32,000 players for fine-grained skill estimation. (**CVPR 2025**)

Graduate Research Assistant

Aug. 2023 – Aug. 2025

UNC-Chapel Hill - Neuroscience Center, Advisor: [Prof. Mark Zylka](#)

Chapel Hill, NC

- Applied 3D CNNs and Video MAEs to automate spontaneous-pain detection in mice.
- Deployed the model to [PainFace](#), a research platform used by 500+ researchers worldwide.

Research Assistant

Oct. 2022 – Aug. 2023

Nanjing Normal University, Advisor: [Prof. Yi Chen](#)

Nanjing, China

- Evaluated a semi-supervised segmentation framework, using feature augmentation to reduce information loss and pixel imbalance. (**Expert Systems with Applications, 2023**)

Research Assistant

Aug. 2022 – May 2023

UNC-Chapel Hill, Advisor: [Prof. Roni Sengupta](#)

Chapel Hill, NC

- Studied motion-robust camera-based vital-sign estimation through dataset curation, augmentation, and rPPG analysis. (**WACV 2024**)

TECHNICAL SKILLS

Programming: Python, C/C++, Swift

ML Frameworks: PyTorch, TensorFlow, OpenCV

Tools & Systems: Git, Docker, Slurm, Linux, Firebase

ACADEMIC SERVICE

Reviewer: CVPR 2026, ECCV 2026, NeurIPS 2026

Workshop Organizer: T4V @ CVPR 2026, LOVEU @ CVPR 2026